

Rebooting the Ocean Ecosystem:

A Concept Proposal for Integrated Microplastic Removal and Ocean Alkalinity Enhancement

Ocean Alkalinization / Ocean Cooling / Marine Microplastics & TRWP Removal

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1. Introduction

1-1. About This Proposal

This proposal addresses both the restoration of the ocean's CO₂ absorption capacity and the microplastic problem simultaneously.

Why “restart” the ocean?

The ocean is Earth's largest sink for excess heat and CO₂.

Restoring this circulation system physically and scientifically is the most efficient and fundamental approach to mitigating global warming.

1-2. Two Critical Bottlenecks Hindering Carbon Absorption

1. Decline in Biological Circulation Functions

Microplastics accumulating in the ocean disrupt the biological activity of plankton and microalgae—the starting point of the marine food chain.

Microplastics may interfere with plankton and marine food-web processes, potentially affecting biological carbon cycling and the ocean's biological pump.

2. Scientific Absorption Limit (Ocean Acidification)

With the rapid rise in atmospheric CO_2 concentrations, seawater pH is decreasing and ocean acidification is accelerating.

The ocean's buffering capacity (alkalinity) may be declining, potentially reducing its capacity to absorb additional CO_2 over the long term.

2. Removal of Marine Plastic

2-1. Estimated Amount of Marine Plastic(1/3)

- Total marine plastic stock: approx. 5–10 million tons

- (1) Floating plastic (surface and shallow layers)

 - Approx. 1–3 million tons

 - Major cause of light obstruction and grazing impacts.

2-1. Estimated Amount of Marine Plastic(2/3)

- Total marine plastic stock: approx. 5–10 million tons

(1) **Floating plastic (surface and shallow layers)**

Approx. **1–3 million tons**

Major cause of light obstruction and grazing impacts.

(2) **Plastic settled on the seafloor**

Approx. **2–5 million tons**

Difficult to measure; future target for recovery by seafloor exploration robots

The least “visible” portion of the total stock.

2-1. Estimated Amount of Marine Plastic(3/3)

(3) **Nanoplastics (unmeasurable portion)**

Approx. **1–2 million tons**

Generated by UV exposure and abrasion

Nearly undetectable with current observation methods

A potential factor in disrupting the biological pump

2-2-1. Potential Restoration of Ocean CO₂ Uptake Capacity

To restore the ocean's ability to absorb CO₂, the following actions are implemented:

- (1) Remove microplastics from surface and shallow layers.
- (2) Ensure sunlight reaches plankton living in the water column.
- (3) Alkalinize seawater.

2-2-2. Potential Restoration of Ocean CO₂ Uptake Capacity

- Use 20–40 m class vessels
- Pump seawater from the surface(1m) and shallow layer(2–5m)
- Filter seawater with UF membranes
 - Remove microplastics
 - Apply electrocoagulation
- Alkalinize treated seawater using a controlled amount of lime.
- Target pH and alkalinity parameters would be determined through site-specific modelling and environmental assessment.
- Disperse the alkalized seawater from the aft section of the vessel.
- No operations will be conducted in the territorial waters of other countries.
(Japan's activities are limited to its own maritime zone.)
- Alkalized seawater will naturally travel to other regions via ocean currents.
- Renewable energy is preferred as the power source for filtration units and pumps.
- Biodiesel is preferred as the fuel for vessels.

2-2-3. Definition of MA Vessel

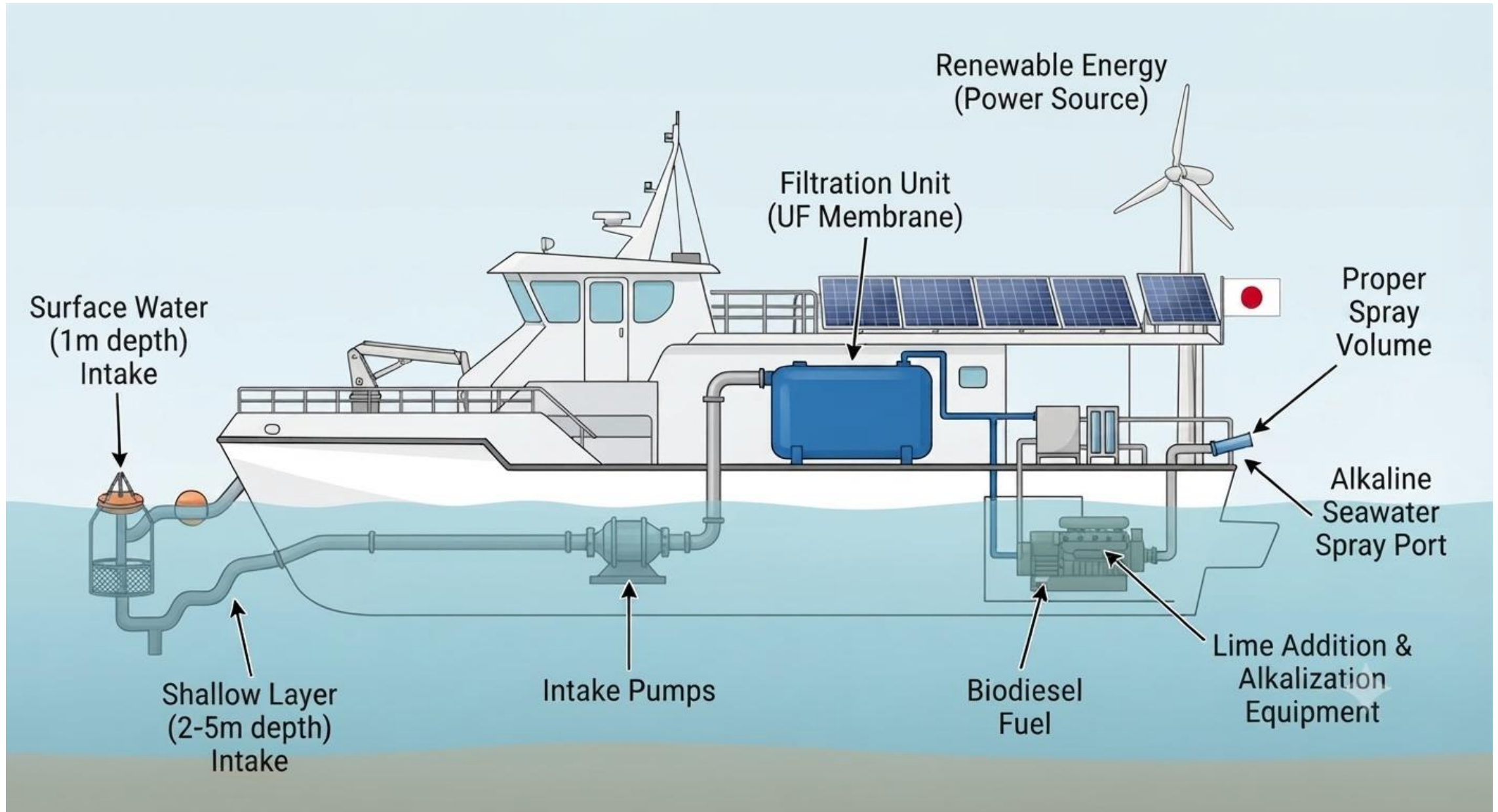
Microplastic & Alkalinization Vessel (MA Vessel)

A specialized vessel designed to:

- **Remove microplastics** using ultrafiltration and electrocoagulation
- **Alkalinize treated seawater using a controlled amount of lime**
- **Disperse treated seawater** to restore CO₂ absorption capacity
- Operate strictly within national maritime zones
- Use **renewable energy** and **biodiesel** to minimize emissions

The MA Vessel is the core operational unit enabling simultaneous microplastic removal and ocean alkalinity enhancement.

2-2-3. Definition of MA Vessel (Image)



2-3-1. Assumptions

- Microplastic concentration varies by region.
- Daily collection capacity per vessel: **5–20 kg/day**.
- Annual operating days: **180 days**, considering weather downtime.

Annual Collection per Vessel

Daily Collection	Operating Days	Annual Collection
5kg/day	180 days	900 kg/year
10kg/day	180 days	1,800 kg/year
12.5kg/day(avg.)	180 days	2,250 kg/year
15kg/day	180 days	2,700 kg/year
20kg/day	180 days	3,600 kg/year

Based on an average collection rate of 12.5 kg/day per vessel

2-3-2. Expected Impact of Multi-Vessel Deployment

- **Annual removal capacity with 100–1,000 vessels:**

 - 100 vessels 225 tons/year (depending on regional microplastic density)

 - 1000 vessels 2,250 tons/year (depending on regional microplastic density)

- **Long-term cumulative removal over 10 years:**

 - 22,500 tons (potentially reducing microplastic concentrations in targeted surface layers)

- **Substantial mitigation of light obstruction and grazing impacts caused by microplastics**

2-3-3. Operation Plan for the MA Vessel

■ Daily Processing Capacity per MA Vessel

Item	Details
Navigation speed (during collection/dispersion)	4 knots (approx. 7 km/h)
Collectable microplastic volume	10–15 kg/day
Alkalinized seawater volume: 300–600 m ³ /day	
Treatable sea surface area	2–4 km ² /day
Dispersion density (per unit area)	0.15 L/m ²

Dispersion density: approximately 0.15 L/m²

2-3-4. Daily Operation Schedule for the MA Vessel

■ Operation Schedule

Item	Details
Departure → Move offshore	1.0H
Morning operations	2.5H (with short breaks as needed)
Lunch break	1.0H
Afternoon operations	2.5H (with short breaks as needed)
Return → Back to port	1.0H
Cleanup & inspection	0.5H
Total	8.5 h (approx. 8.0–9.0 h)
Weather deterioration	Operations suspended; immediate withdrawal

2-3-5. Assigned Ocean Regions

■ Countries Responsible for Each Region

Ocean Region	Responsible Countries
North Pacific	United States, Canada, Russia, Japan
South Pacific	Chile, New Zealand
North Atlantic	United States, Canada
North Atlantic (Eastern Side) / Mediterranean	United Kingdom, France, Spain, Portugal, Ireland, Norway, Italy, Greece, Türkiye, North African coastal countries
South Atlantic	Brazil, Argentina
Indian Ocean	India, Sri Lanka, East African nations, Australia
Western Pacific - East China Sea - Yellow Sea - South China Sea	China, South Korea, Taiwan, Southeast Asian countries

■ Countries that do not face the ocean but generate plastic waste are also expected to cooperate.

2-3-6. Expected Long-Term Impact

If these activities are continued and proven effective through scientific validation, they could contribute to the long-term recovery of the ocean's natural CO₂ uptake capacity.

The magnitude of this effect would require long-term monitoring and independent scientific assessment.

Preliminary conceptual range: 50,000–200,000 t-CO₂/year
(subject to independent scientific validation and field measurements)

If the same annual CO₂ removal (125,000 t-CO₂/year) were achieved through DAC, the cost would be on the order of USD 50–75 million per year, whereas point-source CCUS would still require roughly USD 6–19 million per year.

Appendix A. Other Microplastic Recovery Methods (Outside This Proposal)

Microplastic recovery activities that are considered necessary in addition to the MA Vessel operations:

(1) Remove at Major Accumulation Points

- North Pacific Gyre (Garbage Belt)
- South Pacific Gyre
- North Atlantic Gyre
- South Atlantic Gyre
- Indian Ocean Gyre

(2) Measures to Prevent Outflow from River Mouths

(3) Seafloor Recovery Using Exploration Robots (Future Option)



3. Removal of TRWP

3-1. Why Remove **TRWP**

TRWP (Tire and Road Wear Particles) contain harmful chemical components and are a major contributor to microplastic pollution.

Removing TRWP at the source prevents toxic particles from entering waterways, accumulating in marine ecosystems, and causing long-term environmental damage.

3-2. Key reasons for **TRWP** removal

Major microplastic source:

TRWP are one of the largest sources of microplastic emissions from land-based activities.

Toxic chemical content:

TRWP include zinc, PAHs, and 6PPD-quinone, substances known to be harmful to aquatic organisms.

Biological ingestion:

Their small particle size makes them easily ingested by fish, shellfish, and other marine life.

Persistent accumulation:

TRWP sink and accumulate on the seafloor, where they do not naturally degrade.

Source-level effectiveness:

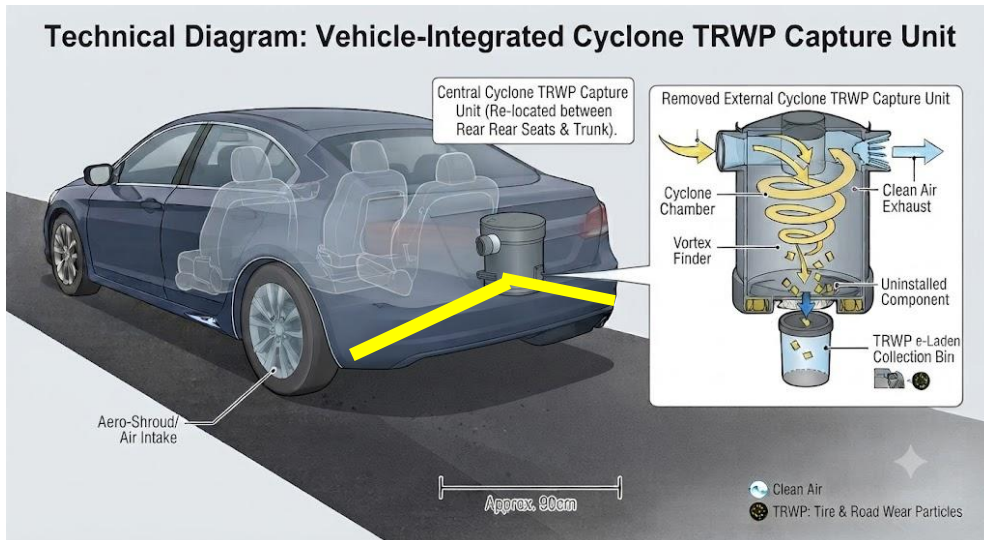
Removing **TRWP** before they enter drainage systems or waterways is far more effective than downstream cleanup.

3-3. **TRWP** Removal Methods

- (1) We propose installing a Cyclone-type dust collector at the rear of the vehicle.
- (2) We suggest installing a Cyclone-type dust collector near major intersections on arterial roads.
- (3) We propose preventing **TRWP** washed into roadside gutters during heavy rain from entering the river system.

3-4-1. Vehicle-Mounted **TRWP** Capture Unit (Draft Specification)

- A **TRWP** capture unit is mounted at the rear of the vehicle.
 - The unit uses a cyclone-type capture system to draw in airborne **TRWP**.
 - Speed-linked control: high suction during high-speed driving, low suction during low-speed driving.
 - The system may be deactivated during rainfall, when airborne resuspension is expected to decrease and runoff becomes the primary transport pathway.
 - Clogging detection is included to ensure stable operation.
 - Collected **TRWP** are stored in a cartridge; the cartridge is replaced when full.
- Waterproof and mud-resistant construction for durability.



Conceptual proposal —
aerodynamic and field testing required

3-4-2. TRWP Capture Units on Major Roadways

Install **TRWP** capture units on **major arterial roads**, particularly near **intersections** and **median strips**.

- Highways
- Primary roads
- Secondary roads

3-4-3. Prevent **TRWP** from Entering Rivers and the Ocean from Roadside Gutters

- Before gutter runoff reaches rivers, **TRWP are removed using microfiltration (MF) membranes.**
- Design capacity should be determined based on local extreme rainfall conditions.
An example design benchmark of 150 mm/hr is assumed.

3-5. Power Source for **TRWP** Capture Units

- **TRWP** capture units installed near intersections on major arterial roads will be powered by **perovskite solar cells combined with non-litium-ion batteries**.
- Perovskite photovoltaic coatings can be applied to **traffic signal poles, road signs, and pedestrian overpasses**.
- The energy storage system will avoid lithium-ion batteries and instead use sodium-ion batteries or other suitable alternatives, subject to safety and lifecycle assessment.

Appendix B. “**TRWP** Removal Methods” 1/3 (Outside This Proposal)

Tire-side

- (1) Low-wear tires (Europe’s ongoing initiative)
 - Adoption of tires designed to reduce abrasion.
 - EU is preparing standards for tire wear rate labeling.

Road-side / Infrastructure Measures

- (2) Road surface optimization
 - Use of asphalt mixes that reduce tire abrasion.
 - High-quality polymer-modified asphalt lowers TRWP generation.

Appendix B. “**TRWP** Removal Methods” 2/3 (Outside This Proposal)

(3) Improved drainage systems

- Side-gutter filters or traps to capture **TRWP** before entering waterways.
- Simple screen filters (e.g., 10–100 μm) may help capture TRWP, depending on particle size distribution and hydraulic conditions.
- Similar measures have already been introduced or are being investigated in parts of Europe.

(Also incorporated into this proposal as a measure to prevent TRWP from entering rivers and the ocean.)

(4) Street sweeping with fine-particle capture

- High-performance sweepers capable of capturing fine particles (down to $\sim 10 \mu\text{m}$).

Appendix B. “**TRWP** Removal Methods” 3/3 (Outside This Proposal)

(5) Traffic flow optimization

- Reducing stop-and-go traffic lowers tire wear.
- Smart signals and smoother traffic patterns reduce **TRWP** generation.

(6) Speed management

- High speeds increase **TRWP** generation.
- Some EU cities use “abrasion-aware” speed limits in sensitive zones.

(7) Roadside vegetation strips

- Vegetation traps airborne **TRWP** before they disperse.
- Proven in Scandinavian studies.

A combined portfolio of measures is likely to be more effective than any single intervention.

4. Summary

Climate and Ocean Recovery Actions

- Remove microplastics from the ocean surface and upper layers.
 - Increase ocean alkalinity (raise pH by 0.05–0.15).
 - Prevent **TRWP** from entering rivers and the ocean.
 - Aims to contribute to the restoration of the ocean's natural **CO₂** absorption capacity.
 - May contribute to reducing excess heat accumulation in the ocean over the long term.
 - In the long term, reducing atmospheric **CO₂** can contribute to a reduction in radiative forcing and global warming.
- The timescale and magnitude of this effect require scientific modelling.

We cannot wait until global **CO₂** emissions reach net-zero in 2050.

Early action is essential.

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I am entrusting this to a promising young person.

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■ Revision History

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